

PALE MERIDIAN - SCALE & GAMEPLAY READABILITY BENCHMARK V2

Result image set: every render from scale_23 / scale_25 / scale_27 / scale_30. Deterministic code-generated pixel art (Python + integer raster, PIL save only).

Camera: tilted orthographic top-down (CrossCode / UNSIGHTED family), pitch 36 deg, azimuth 0, wall lean 2 deg, no perspective, no 3D. $sx = x*S$; $sy = \text{round}(\sin 36^\circ * y * S - \cos 36^\circ * z * S)$.

Building: MERIDIAN DEPOT 9 - frontier sci-fi service depot, ~31 m wide: forward annex (own entrance) + main hall (cargo door, personnel door, window band, wall-mounted equipment) + set-back wing (side entrance) + rear tech block with stack + corner pilaster; 4 entrances, 5 masses, parapet roofs with HVAC / vent / skylight / hatch / cable tray; service yard with combat test pad.

Gameplay objects on the pad: human reference 1.8 m (neutral silhouette), large crate 1.2 m, small crate 0.6 m, industrial terminal 1.35 m, generator 2.4 m, industrial container 3.0 m, cover barricade 1.8x1.3 m, wall-mounted unit, floor equipment, lamp post 4.5 m, debris cluster. Average object spacing 2.41 m.

scale dir	px/m	world window 480x270	human px	personnel door px	generator px	container px	cover px	detail gen
scale_23	23	20.87 x 11.74	41	25x49	55x37	69x37	41x30	0
scale_25	25	19.20 x 10.80	45	28x54	60x40	75x40	45x32	1
scale_27	27	17.78 x 10.00	49	30x58	65x43	81x43	49x35	2
scale_30	30	16.00 x 9.00	54	33x64	72x48	90x48	54x39	3

Reading order per scale: GAME (what the player sees) -> COMBAT (pad readability) -> CLOSE (depth grammar) -> OVERVIEW (architecture) -> SHEET -> crops (native-pixel detail). Higher px/m = higher PRODUCTION RESOLUTION (authored detail tiers, physical periods in metres), NOT camera zoom: the world window shrinks while every object gains pixels. Main result (REPORT.md): 27 px/m @480x270 is the balance point; 30 px/m @640x360 regains the wide window with ~1.7x pixels per object; 23 px/m stays the lower bound.

SCALE 23 px/m (A)

pixels_per_meter / projection	23 tilted orthographic, pitch 36 deg, azimuth 0 deg, lean 2 deg
logical resolution	game/combat 480x270, close 320x220, overview 828x285
world window (game)	20.87 x 11.74 m
building bbox (overview px)	713 x 255
human / doors	human 41 px personnel 25x49 cargo 83x74
objects px	generator 55x37 container 69x37 cover 41x30 crate L 28x25 terminal 13x31 lamp post 104 px
pixel construction	gen 0 recess 3 px shadow rows 4 cast step 2 px eave 3 px chamfer 3 px stencil scale 2
metrics	cast-shadow px 19283 unique colors 45 silhouette line px 2096
review (1-5)	object 4 arch 4 material 4 depth 4 mass 4 gameplay space 4 detail capacity 3 action 4

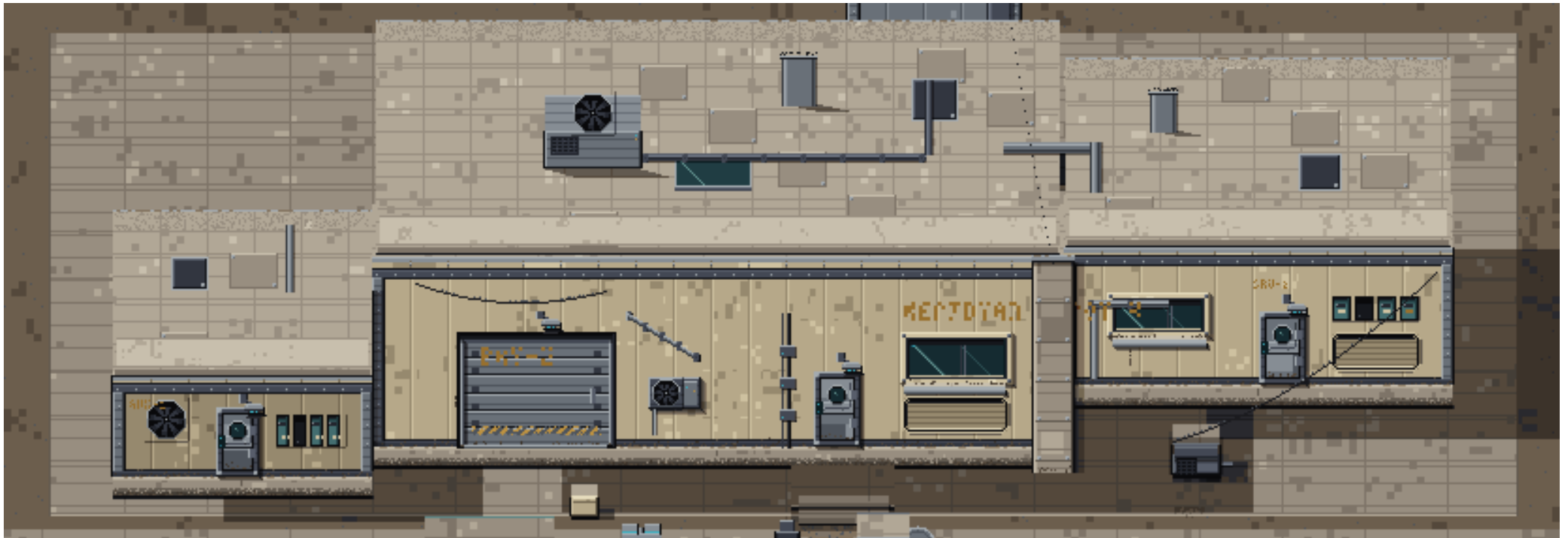
Verdict: baseline of the new range: window 20.9x11.7 m at 480x270; objects read as forms with function; secondary forms 3 px; some micro-detail still at the edge of legibility



combat.png - COMBAT SPACE VIEW 480x270 - camera centred on the combat test pad (anchor 10.6, 6.2, 1.3 m): human reference, cover, crates, terminal, generator, passage between cover and large crate.



close.png - CLOSE VIEW 320x220 - personnel door + entrance platform + human reference + wall equipment; door depth grammar: opening -> frame -> recess -> leaf -> threshold -> local shadow.



overview.png - BUILDING OVERVIEW - fixed world window 36x20 m (anchor 9.5, -3.5, 3.0 m): annex + hall + wing + rear tech block + corner pilaster, roofs with machinery, service yard.



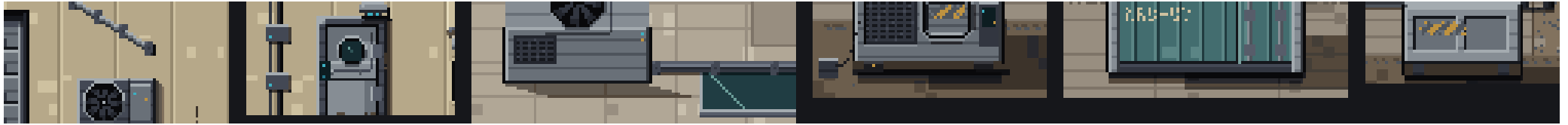
sheet.png - PER-SCALE SHEET - game + combat views and 1x crops of human / personnel door / generator.



crops_detail.png - DETAIL CROPS (same world window per subject, native px): door, window, generator, crate, container, cover, terminal.



crops_gameplay.png - GAMEPLAY CROPS (native px): human, door, cover, generator, crate, object spacing / passage.



crops_inspect.png - INSPECTION CROPS (native px): wall, door, roof, generator, container, cover.

SCALE 25 px/m (B)

pixels_per_meter / projection	25 tilted orthographic, pitch 36 deg, azimuth 0 deg, lean 2 deg
logical resolution	game/combat 480x270, close 320x220, overview 900x310
world window (game)	19.20 x 10.80 m
building bbox (overview px)	775 x 278
human / doors	human 45 px personnel 28x54 cargo 90x80
objects px	generator 60x40 container 75x40 cover 45x32 crate L 30x28 terminal 14x34 lamp post 112 px
pixel construction	gen 1 recess 4 px shadow rows 4 cast step 2 px eave 3 px chamfer 3 px stencil scale 2
metrics	cast-shadow px 20808 unique colors 46 silhouette line px 2092
review (1-5)	object 4 arch 4 material 4 depth 4 mass 4 gameplay space 4 detail capacity 4 action 4

Verdict: recesses and shadow bands gain a pixel row; cover and crates start to read as physical volumes; window 19.2x10.8 m still keeps the whole combat pad in frame



game.png - GAME VIEW 480x270 - combat camera in front of the hall entrance (anchor 9.8, 3.6, 1.7 m). Foreground props, combat pad, facade fragment; the building never fits whole at 30 px/m.



combat.png - COMBAT SPACE VIEW 480x270 - camera centred on the combat test pad (anchor 10.6, 6.2, 1.3 m): human reference, cover, crates, terminal, generator, passage between cover and large crate.



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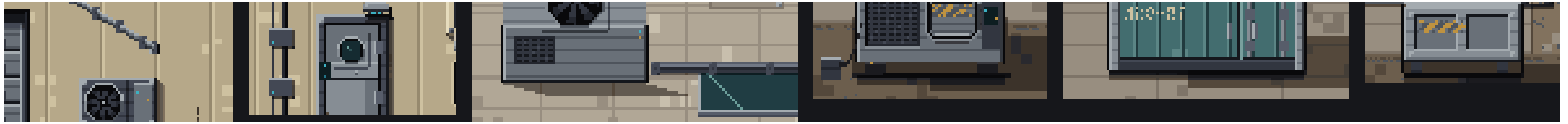
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crops_detail.png - DETAIL CROPS (same world window per subject, native px): door, window, generator, crate, container, cover, terminal.



crops_gameplay.png - GAMEPLAY CROPS (native px): human, door, cover, generator, crate, object spacing / passage.



crops_inspect.png - INSPECTION CROPS (native px): wall, door, roof, generator, container, cover.

SCALE 27 px/m (C)

pixels_per_meter / projection	27 tilted orthographic, pitch 36 deg, azimuth 0 deg, lean 2 deg
logical resolution	game/combat 480x270, close 320x220, overview 972x335
world window (game)	17.78 x 10.00 m
building bbox (overview px)	837 x 300
human / doors	human 49 px personnel 30x58 cargo 97x86
objects px	generator 65x43 container 81x43 cover 49x35 crate L 32x30 terminal 15x36 lamp post 122 px
pixel construction	gen 2 recess 4 px shadow rows 4 cast step 2 px eave 4 px chamfer 3 px stencil scale 2
metrics	cast-shadow px 21694 unique colors 49 silhouette line px 2006
review (1-5)	object 5 arch 4 material 5 depth 5 mass 5 gameplay space 4 detail capacity 4 action 5

Verdict: door/window/generator depth grammar fully legible at 1x; human 49 px gives a confident scale anchor; window 17.8x10.0 m - pad plus facade fragment, best balance



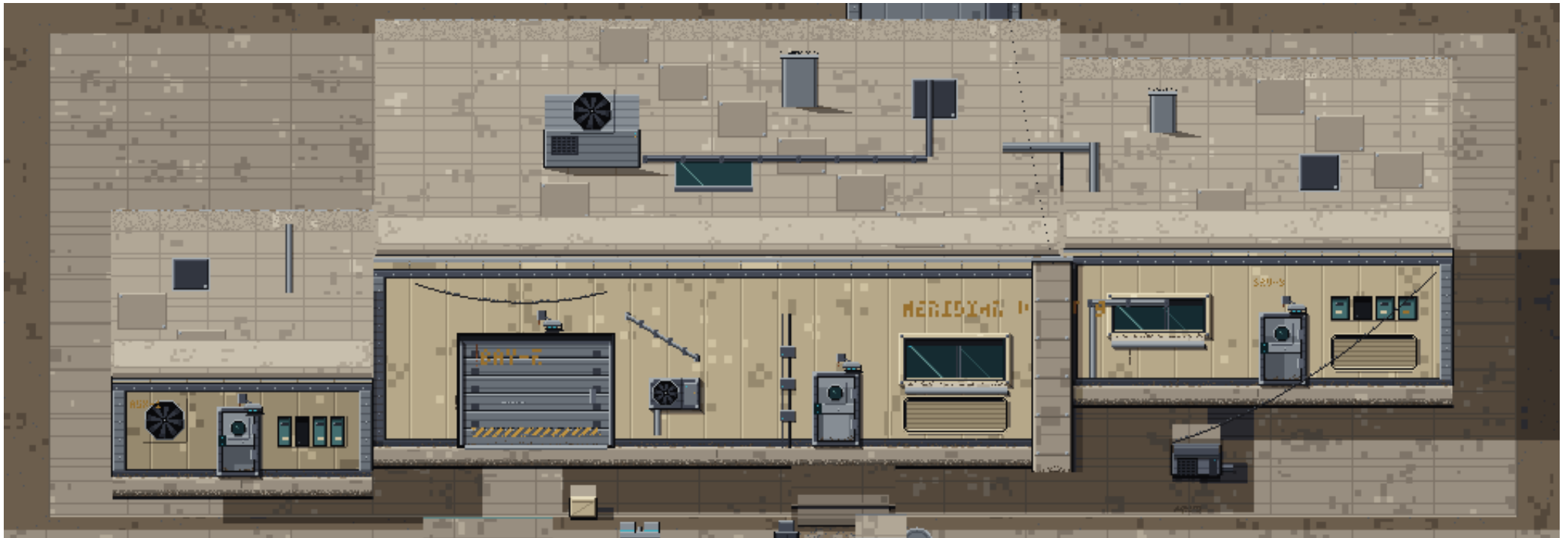
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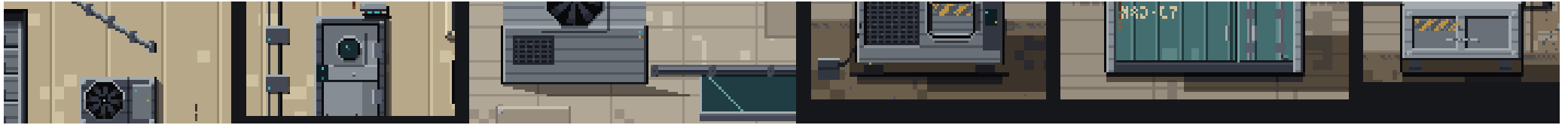
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SCALE 30 px/m (D)

pixels_per_meter / projection	30 tilted orthographic, pitch 36 deg, azimuth 0 deg, lean 2 deg
logical resolution	game/combat 480x270, close 320x220, overview 1080x372
world window (game)	16.00 x 9.00 m
building bbox (overview px)	930 x 333
human / doors	human 54 px personnel 33x64 cargo 108x96
objects px	generator 72x48 container 90x48 cover 54x39 crate L 36x33 terminal 16x40 lamp post 135 px
pixel construction	gen 3 recess 4 px shadow rows 5 cast step 2 px eave 4 px chamfer 3 px stencil scale 2
metrics	cast-shadow px 22468 unique colors 48 silhouette line px 2058
review (1-5)	object 5 arch 4 material 5 depth 5 mass 5 gameplay space 3 detail capacity 5 action 4

Verdict: maximum presence and pixel budget; window 16.0x9.0 m at 480x270 makes the combat pad overflow the frame - strong for close action, tight for navigation; at 640x360 the same scale regains the 23px/m world window with far more pixels per object



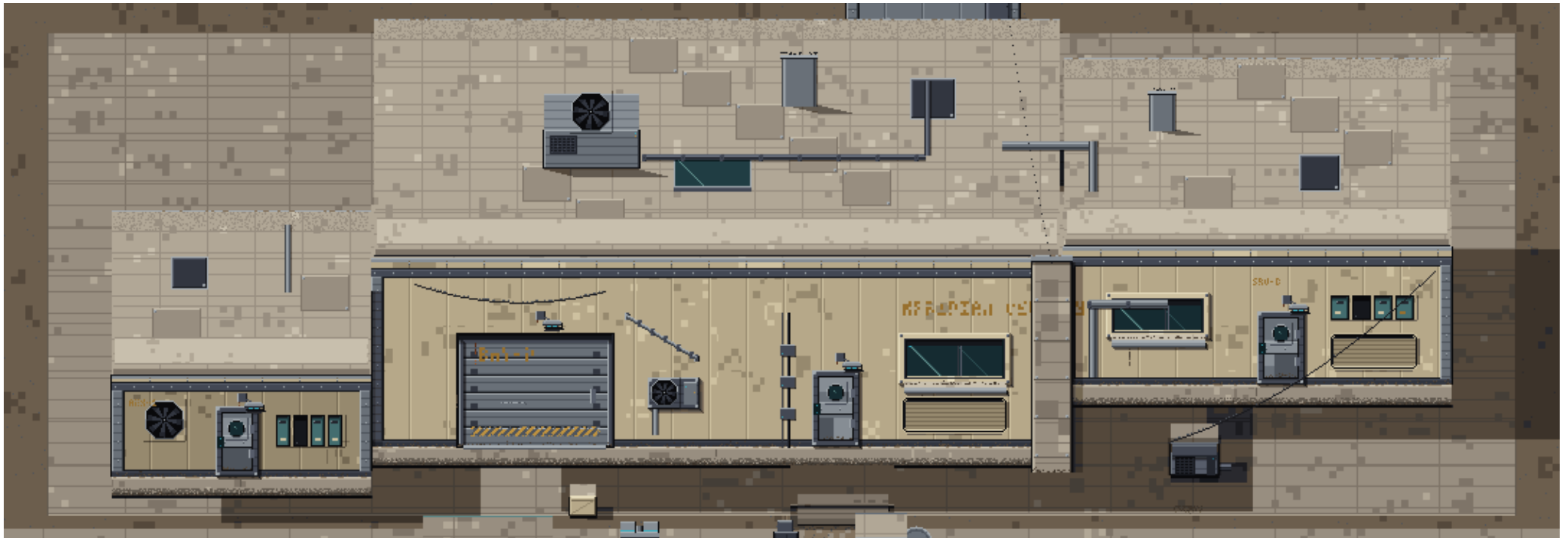
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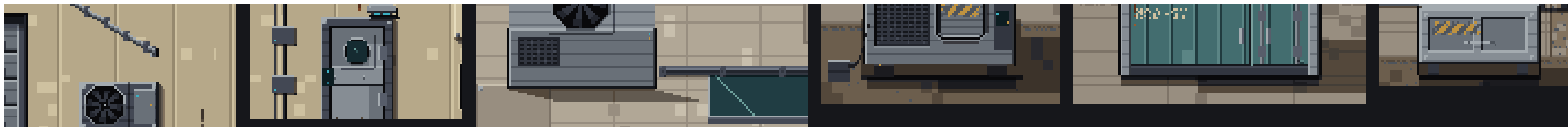
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